

Chapter 8: Teratology, Teratogens, and Fetotoxic Agents — Revision Notes

Williams Obstetrics 26e, pp. 383–434 of the PDF. High-yield numbers are in **bold**. The five tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 4 figures are included from the PDF.

Big picture: Major birth defects affect **2–3%** of newborns, but **<1%** of all birth defects are caused by medications. Few drugs are true teratogens, and for almost every one a safer alternative exists. Counseling should use **absolute risk**, not odds ratios.

1. Background and Definitions

- **2–3%** of newborns have a major congenital abnormality detectable at birth.
- **80%** of birth defects have **no obvious cause**. Of those with a known cause, **~95% are chromosomal or genetic**.
- FDA: **<1% of birth defects are caused by medications**.
- Medication use is common:
 - Average **2–3 medications per pregnancy**; **70%** use a medication in the 1st trimester.
 - **65%** fill ≥ 1 prescription in pregnancy (Canada).
 - Of 290 new drugs approved in 2010–2019, only **11%** had human pregnancy data.

Table 8-1. Selected Teratogenic and Fetotoxic Agents

A–L	L–P	R–W
Acitretin	Leflunomide	Radioactive iodine
Alcohol	Lithium	Ribavirin
Angiotensin-converting enzyme inhibitors	Marijuana	Sulfonamides
Angiotensin-receptor blockers	Mercury	Tamoxifen
Bexarotene	Methamphetamine	Tetracyclines
Carbamazepine	Methimazole	Thalidomide and analogues ^b
Cocaine	Methotrexate	Tobacco
Corticosteroids	Mycophenolate	Toluene
Cyclophosphamide	Nitrofurantoin	Topiramate
Endothelin-receptor antagonists ^a	Non-steroidal anti-inflammatory drugs	Trastuzumab
Fluconazole	Paroxetine	Valproic acid
Isotretinoin	Phenobarbital	Warfarin
Lead	Phenytoin	

^aIncludes ambrisentan, bosentan and macitentan. ^bIncludes lenalidomide and pomalidomide.

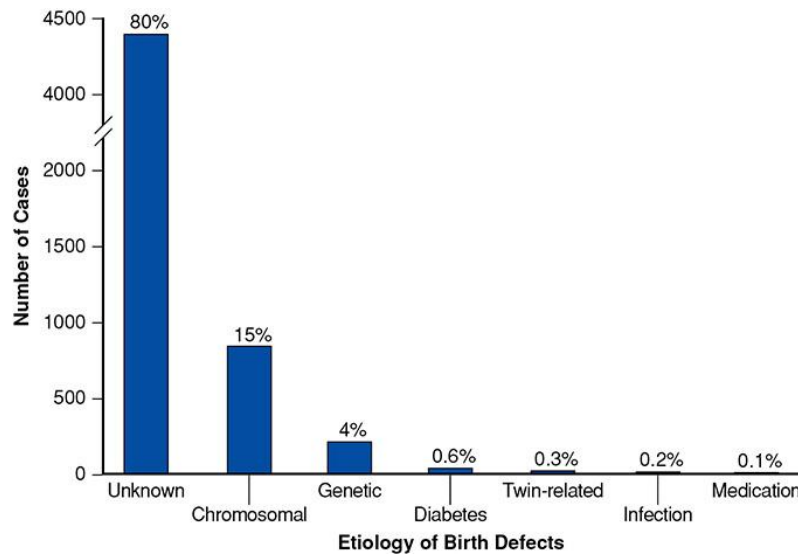


Figure 8-1. Causes of 5504 birth defects (270,878 births): unknown 80%, chromosomal 15%, genetic 4%, diabetes 0.6%, medication only 0.1%.

2. Teratology

Term	Definition
Teratogen	Any agent acting during embryonic or fetal development that produces a permanent alteration of form or function . Strictly, it causes structural abnormalities.
Hadegen (after Hades)	Interferes with normal maturation and function of an organ
Trophogen	Alters growth

- Teratogens include drugs/chemicals, physical agents (**heat, radiation**), maternal metabolites (**diabetes, PKU**) and infections (**CMV**).
- Hadegens and trophogens act mostly in the fetal period or after birth → they are **fetotoxins** rather than teratogens.

3. Criteria for Determining Teratogenicity (Shepard)

Table 8-2. Criteria for Determining Teratogenicity

Essential criteria

1. Careful delineation of clinical cases, particularly if there is a specific defect or syndrome
2. Proof that exposure occurred at a critical time during development (see Fig. 7-2)
3. Consistent findings by **at least two epidemiological studies** with:
 - a. exclusion of bias
 - b. adjustment for confounding variables
 - c. adequate sample size (power)
 - d. prospective ascertainment if possible
 - e. **relative risk (RR) ≥ 3.0** ; some recommend $RR \geq 6.0$

or for a rare environmental exposure associated with a rare defect, **at least three reported cases**. This is easiest if the defect is severe.

Ancillary criteria

1. The association is biologically plausible

2. Teratogenicity in experimental animals is important but **not essential**
3. The agent acts in an unaltered form in an experimental model

Key tenets

- **Defect completely characterized**, ideally by a geneticist/dysmorphologist. Causation is easiest to prove when a **rare exposure** → **rare anomaly**, ≥ 3 cases, severe defect.
- **Agent must cross the placenta** in sufficient quantity. Transfer depends on maternal metabolism, protein binding, molecular size, charge, lipid solubility and placental metabolism (**cytochrome P450**). The early placenta has a thick membrane that slows diffusion.
- **Exposure in a critical period:**

Period	Timing	Effect of an insult
Preimplantation ("all or none")	First 2 wk after fertilization	Many cells damaged → embryonic death ; few damaged → compensation, normal development
Embryonic	2nd–8th wk postconception	Organogenesis → most crucial period for structural malformations
Fetal	>8 wk postconception	Maturation and functional development; some organs remain vulnerable

- **Biological plausibility** supports the link (defects and drug use are both common, so they can be temporally but not causally related).
- **Consistent epidemiology**: ≥ 2 high-quality studies; **RR ≥ 3.0** needed. Pitfalls: recall bias, poor reporting, confounders (dose, other drugs, maternal disease).
- **Animal data**: not obligatory; for litigation, **human data are required**.

Bendectin — the cautionary tale

- **Doxylamine + pyridoxine** ($\pm$ dicyclomine) for nausea and vomiting; used by >30 million women.
- Anomaly rate **3%** = background rate. Still, lawsuits forced withdrawal in **1983** → **hospitalization for hyperemesis doubled**.
- Remarketed as **Diclegis**, FDA-approved in **2013**.

4. Studies in Pregnant Women

- Problems:
 - **Animal studies are necessary but insufficient** (thalidomide was harmless in several animal species → **phocomelia** in thousands of children).
 - Drugs are rarely approved for pregnancy indications; pregnant women are routinely excluded from research.
 - Pregnancy physiology changes drug levels (volume of distribution, CO, GI absorption, hepatic metabolism, renal clearance).

Study design	Strength	Weakness
Case reports/series ("astute clinician model")	Found rubella (Gregg, 1941), thalidomide, alcohol. Need proven exposure at a critical time + ≥3 cases	No control group ; misses uncommon exposures, nonspecific defects or low-penetrance effects
Case-control	Efficient for rare outcomes ; generates hypotheses	Recall bias ; confounding by indication ; power to detect trivial differences. Associations may be wrong unless OR >3–4
Cohort (e.g., Medicaid, insurance claims)	Exposed vs unexposed	Needs very large samples; can't adjust for confounders such as indication
Pregnancy registries	Prospective enrollment of exposed pregnancies (FDA lists registries for 115 drugs)	No control group ; need baseline prevalence (e.g., Metropolitan Atlanta Congenital Defects Program, 1967)

National Birth Defects Prevention Study (NBDPS)

- Population-based **case-control** study, **1997–2013**, 10 states; **~32,000 cases**, **~12,000 controls**; included live births, stillbirths and terminations.
- Found small associations with antibiotics, antidepressants, antiemetics, antihypertensives, asthma drugs, **NSAIDs, opioids**; also secondhand smoke, pesticides, nitrogen oxide (traffic pollution).
- **Pregestational diabetes** was associated with **46 of 50** abnormalities studied.
- **Limitations:**
 - Interviews **6 wk to 2 yr** after delivery → recall bias (25% couldn't remember which antibiotic).
 - Only $\frac{2}{3}$ participated → selection bias.
 - No record review → no dose-response.
 - **No adjustment for multiple comparisons** (antibiotics: 43 comparisons, 4 significant; 2 expected by chance).
 - Absolute risks often as low as **1 per 1000** exposed.

5. Counseling for Medication Exposure

- Ask about medication and illicit drug use at every preconceptional and prenatal visit.
- Women **underestimate background risk** and **overestimate drug risk**:
 - Utah: of 5500 major defects in >270,000 births, only **4** were due to medication.
 - **$\frac{1}{4}$** of women exposed to nonteratogenic drugs thought they had a **25%** risk of anomalies.
- Good counseling ↓ anxiety and **may avert termination**.
- **Sources:** the manufacturer's prescribing information (recommended first), PubMed, **Reprotox, TERIS, Shepard's Catalog; LactMed** for breastfeeding.
- Weigh dose, route, timing, other drugs, the underlying disease — and **the risk of not treating**.

Labeling requirements

- **1979:** FDA letter categories **A, B, C, D, X** → led to over-reliance on the letter.
- **2015:** new **Pregnancy and Lactation Labeling** replaced the letters with a narrative summary of risks, clinical considerations and data. Also includes registry info, a lactation subsection, a reproductive-potential section, and the **background risk of birth defects and miscarriage**.

Table 8-3. Food and Drug Administration Pregnancy Labeling Requirements

Label Subsection	Categories	Selected Specific Content
8.1 Pregnancy	Pregnancy exposure registry	Presence of a registry and contact information
	Risk summary	Structural abnormalities, embryo-fetal and/or infant mortality information, functional impairments, and alterations to growth
		Human data, animal data, and pharmacology information as an integrated summary
		Incidence of adverse outcomes, including dose effects, exposure duration, and gestational age
	Clinical considerations	Disease-associated maternal and embryo-fetal risks; dose adjustments (pregnancy or postpartum); maternal, fetal or neonatal adverse reactions; effects on labor or delivery
	Data	Content from human and animal studies and their findings and limitations
8.2 Lactation	Risk summary	Presence of a drug or its metabolite(s) in human milk, effects on the breastfed child, and effects on milk production
	Clinical considerations	Summary of available data and information about minimizing exposure
	Data	Includes content from human and animal studies that inform clinical considerations
8.3 Males and females of reproductive potential	Pregnancy testing	Testing requirement or recommendation before, during, or after drug use
	Contraception	Contraception requirement or recommendation before, during, or after drug use
	Infertility	Effects on fertility, potential reversibility, mutagenesis data, and pre-implantation loss effects

From Food and Drug Administration, 2020b.

Presenting risk information

- Cover the fetal risk of the drug **and** the risk of the disease itself **and** of not treating it.
- **Framing matters:** "2% chance of a malformation" sounds worse than "**98% chance of an unaffected baby**".
- Use **absolute or attributable risk** (prevalence in exposed – unexposed), not odds ratios.
 - Example: oral corticosteroids + cleft lip = "tripled risk" sounds alarming, but it is **1 → 3 per 1000**, or a **99.7%** chance of no cleft.
- **Low-risk teratogen** = defects in **<10 per 1000** exposures; close to background, so usually not a reason to stop needed treatment.
- **Background risk of a major anomaly ≈ 3%** for all women. A confirmed teratogen usually adds only **1–2%**, or at most doubles or triples it.

6. Teratogenic and Fetotoxic Agents — General

- Relatively few drugs are major teratogens, and **a safer alternative almost always exists**.
- Take any medication only when clearly needed.
- **Detailed sonography** is generally indicated after exposure to a major teratogen during the **embryonic period**.

Alcohol

- **Leading cause of preventable developmental disability** worldwide.

- US: **10%** of pregnant women drink; ~10% of these binge in the 1st trimester (1% in the 2nd–3rd).
- Described by **Lemoine (1968)** and **Jones (1973)**.
- **Fetal alcohol spectrum disorder (FASD)** — umbrella for **5 conditions**:
 1. Fetal alcohol syndrome (FAS)
 2. Partial FAS
 3. Alcohol-related birth defects
 4. Alcohol-related neurodevelopmental disorder
 5. Neurobehavioral disorder associated with prenatal alcohol exposure
- **Prevalence**:
 - FASD >1% in 76 countries; **1 in 13** women who drink in pregnancy has a child with FASD.
 - FAS up to **1%** of US births; FASD in **2–5%** of school-aged children.
- Other associations: **omphalocele, gastroschisis** (periconceptional use); **binge drinking** → highest risk for defects and **stillbirth**.
- **No sonographic criteria** for prenatal diagnosis. Parkland: detailed scan if exposure criteria are met ± 3rd-trimester growth scan.
- Susceptibility is modified by genetics, nutrition, maternal disease and age.
- **No amount of alcohol is safe in pregnancy** (ACOG, CDC, AAP).

Table 8-4. Criteria for Prenatal Alcohol Exposure, Fetal Alcohol Syndrome, and Alcohol-Related Birth Defects

Documented prenatal alcohol exposure — one or more required

1. **≥6 drinks per week for ≥2 weeks**
2. **≥3 drinks per occasion for ≥2 occasions**
3. Risk identified with a validated screening questionnaire
4. Laboratory testing indicating alcohol intoxication or positive alcohol-exposure biomarker
5. Documentation of an alcohol-related legal or social problem

Fetal alcohol syndrome diagnostic criteria — ALL required

1. **Dysmorphic facial features (≥2 required)**:
 - a. Short palpebral fissures
 - b. Thin vermilion border of the upper lip
 - c. Smooth philtrum
2. Prenatal and/or postnatal **growth impairment, ≤10th percentile**
3. Abnormal brain growth, morphogenesis, or physiology (≥1 required):
 - a. **Head circumference ≤10th percentile**
 - b. Structural brain abnormalities
 - c. Recurrent nonfebrile seizures
4. **Neurobehavioral impairment** (defined as **>1.5 SD below mean**):
 - a. Child <3 years: developmental delay
 - b. Child ≥3 years: global cognitive impairment, or cognitive deficit in at least 1 neurobehavioral domain, or behavioral deficit in at least 1 domain

Alcohol-related birth defects

System	Defects
Cardiac	Atrial or ventricular septal defect, aberrant great vessels, conotruncal heart defects
Skeletal	Radioulnar synostosis, vertebral segmentation defects, joint contractures, scoliosis
Renal	Aplastic or hypoplastic kidneys, dysplastic kidneys, horseshoe kidney, ureteral duplication
Eyes	Strabismus, ptosis, retinal vascular abnormalities, optic nerve hypoplasia
Ears	Conductive or neurosensory hearing loss

SD = standard deviation.

Memory hook: FAS = "**3 faces + growth + brain + behavior**" — short **P**alpebral fissures, thin **U**pper lip, smooth **P**hiltrum ("PUP"), ≥ 2 of 3.

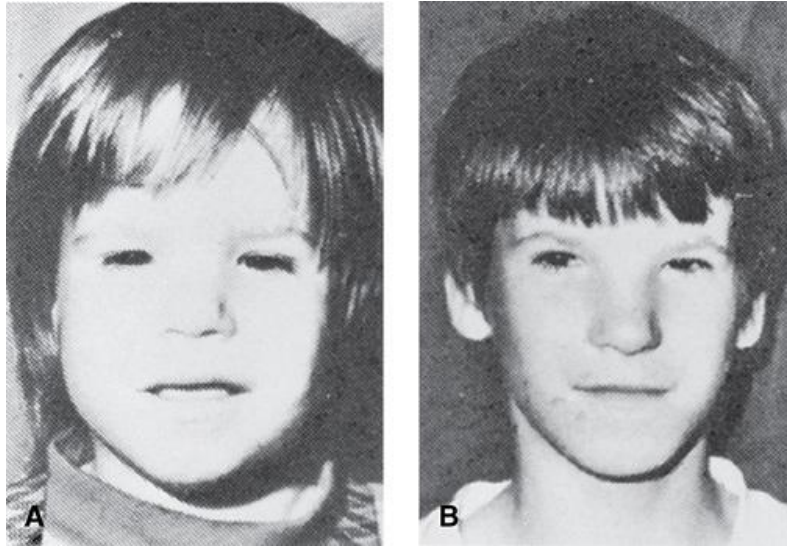


Figure 8-2. Fetal alcohol syndrome at 2½ and 12 yr: short palpebral fissures, epicanthal folds, flat midface, hypoplastic philtrum and thin upper lip persist.

7. Antiepileptic Drugs

- Most common anomalies: **orofacial clefts, cardiac malformations, neural-tube defects (NTDs)**.
- **Valproic acid = highest risk:**
 - NAAED registry: major malformations in **9%** after 1st-trimester exposure, including **4% NTDs**.
 - Exposed children have **lower IQ** and poorer cognitive development than with other AEDs.
- Malformation rate vs untreated epilepsy (monotherapy):

Drug	Relative malformation risk
Carbamazepine, phenytoin	2×
Phenobarbital	3×
Topiramate	4×
Polytherapy	Risk roughly doubled
Levetiracetam, lamotrigine	No increase (levetiracetam 2% = population rate; >1500 lamotrigine pregnancies, no ↑; no oral clefts in 208 Israeli cases)

- **Fetal hydantoin syndrome** (older AEDs): **hypertelorism, low nasal bridge, midface hypoplasia; hypoplastic distal phalanges and nails**; growth impairment; intellectual disability.
- Enroll exposed women in the **NAAED Pregnancy Registry**.

8. ACE Inhibitors and Angiotensin-Receptor Blockers

- **ACE-inhibitor fetopathy:** fetal kidney development depends on the fetal renin–angiotensin system.
 - Fetal hypotension → **renal hypoperfusion** → **ischemia and anuria**.
 - → **FGR, calvarial maldevelopment; oligohydramnios** → **pulmonary hypoplasia and limb contractures**.
- Exposure **beyond 20 wk:** fetopathy risk **30% with ARBs** vs **3% with ACE inhibitors**.
- **1st-trimester (embryo) toxicity largely disproven:** an early Tennessee Medicaid study (2–3× cardiac/CNS defects) was not confirmed in cohorts of **460,000** and **1.3 million** pregnancies.
- → Women with **inadvertent 1st-trimester exposure can be reassured**, but **avoid ACEi/ARBs in pregnancy** — use other antihypertensives.

9. Antifungals

- **Fluconazole, chronic high dose (400–800 mg/d), 1st trimester** → pattern like **autosomal recessive Antley-Bixler syndrome**: oral clefts, abnormal facies, cardiac, skull, long-bone and joint anomalies.
- **Low dose (150–300 mg total) for vulvovaginal candidiasis:**
 - No overall rise in birth defects (small cardiac risk not excluded).
 - Denmark: **3× tetralogy of Fallot** (3 → 10 per 10,000), but a later study of >37,000 exposures found **no cardiac association**.
- Parkland: **no** detailed scan or fetal echo after low-dose fluconazole.

10. Antiinflammatory Agents

NSAIDs (aspirin, ibuprofen, indomethacin) — inhibit prostaglandin synthesis

- **≥20%** take them in the 1st trimester; **not a major risk factor for birth defects**.
- **Late pregnancy — indomethacin:**
 - **Constriction of the ductus arteriosus** → **pulmonary hypertension**; more likely with 3rd-trimester use **>72 h**.
 - ↓ fetal urine → **oligohydramnios**.
 - Tocolysis → **1.5×** risk of BPD, IVH and NEC.
- **FDA: avoid NSAIDs from ≥20 wk** (oligohydramnios/neonatal renal problems).
 - If needed for **>48 h**, consider ultrasound of amniotic fluid.
 - Oligohydramnios usually resolves **within 6 days** of stopping, but some infants died of renal failure.
- **Low-dose aspirin ≤100 mg/d is safe** (no ductal constriction). Avoid **high-dose** aspirin, especially in the 3rd trimester.

Leflunomide (pyrimidine-synthesis inhibitor, for RA)

- **Contraindicated** in pregnancy (animal: hydrocephalus, eye/skeletal anomalies, embryo death).
- Active metabolite **teriflunomide persists up to 2 years**.
- If pregnant (or planning, even after stopping) → **accelerated elimination with cholestyramine or activated charcoal**.
- Human data are reassuring: >500 exposures, major defects **3%** (= population); no ↑ miscarriage.

11. Antimicrobial and Antiviral Drugs

- Most commonly used antimicrobials have **no embryo-fetal safety concerns**.

Drug	Concern	Practical point
Nitrofurantoin	NBDPS: 2× cleft lip (still 998/1000 unaffected). Case-control: 3× hypoplastic left heart (<1/1000) — not seen in cohort studies	Contraindicated with G6PD deficiency (hemolysis). Fine in 2nd–3rd trimester; 1st trimester if no suitable alternative
Sulfonamides (TMP-SMX)	NBDPS: 5× esophageal atresia or diaphragmatic hernia (~1/1000); not confirmed in >7500 exposures	OK in 1st trimester if no alternative. Contraindicated with G6PD deficiency . Displace bilirubin → theoretical neonatal hyperbilirubinemia near preterm delivery (not seen in Danish data)
Tetracyclines	Yellow-brown staining of deciduous teeth if used after 25 wk ; no ↑ caries	Doxycycline preferred (less calcium chelation; no ↑ defects or staining)
Ribavirin (hepatitis C)	Animal teratogen at low doses (skull, palate, eye, skeleton, GI). Half-life 12 days ; persists in tissues	Two forms of contraception + monthly pregnancy tests during and for 6 months after. Also contraindicated in men whose partners are pregnant

12. Antineoplastic Agents

- First-trimester chemotherapy exposure: **14%** major malformations (300 pregnancies).

Agent	Key facts
Cyclophosphamide (alkylating)	↑ pregnancy loss; skeletal, limb, cleft palate, eye defects. Major anomaly risk 18% . Growth and developmental delay. Health-care workers exposed → ↑ miscarriage
Methotrexate (folic acid antagonist)	Fetal methotrexate-aminopterin syndrome : craniosynostosis with " cloverleaf " skull, wide nasal bridge, low-set ears, micrognathia; CNS, cardiac and limb defects; intellectual impairment. Most vulnerable at 8–10 wk postconception and ≥10 mg/week . Persists in the body → preconceptional exposure is not risk-free . The 50 mg/m² ectopic/abortion dose exceeds the threshold → conotruncal heart defects reported after inadvertent treatment of an intrauterine pregnancy
Tamoxifen (SERM)	DES-like defects in rodents (vaginal adenosis). Human: abnormality in 13% of 167 (ambiguous genitalia, craniofacial)
Trastuzumab (anti-HER2 antibody)	Not a malformation risk, but oligohydramnios sequence → pulmonary hypoplasia, renal failure, skeletal anomalies, neonatal death. Surveillance if exposed in pregnancy or within 7 months before conception . Same for ado-trastuzumab emtansine

13. Endothelin-Receptor Antagonists

- **Bosentan, ambrisentan, macitentan** (pulmonary arterial hypertension).
- Endothelin signaling is needed for **neural-crest development** → craniofacial and **cardiac outflow tract** defects in animals. No human data.
- Available only through **restricted access programs**: contraception + **monthly pregnancy testing**.

14. Immunosuppressants

- **Corticosteroids**:
 - Systemic use → **3× orofacial clefts**, absolute risk only **3 per 1000**; no overall rise in major malformations → **not a major teratogen**.

- **Prednisone/methylprednisolone** are lower risk: prednisolone is inactivated by placental **11 β -hydroxysteroid dehydrogenase 2**, so less reaches the fetus.
- **Mycophenolate mofetil / mycophenolic acid** (IMP dehydrogenase inhibitor) — **potent teratogen**:
 - Continued past the 1st trimester: **30% birth defects + 30% miscarriage**. >20% of liveborns have major anomalies.
 - **Mycophenolate embryopathy**: **microtia, auditory canal atresia**, clefts, **coloboma**/eye anomalies, short fingers with hypoplastic nails, cardiac defects.
 - Prescribing requires a **REMS** (FDA-mandated Risk Evaluation and Mitigation Strategy).

15. Environmental Metals

Lead

- Crosses the placenta by passive diffusion → **FGR** and childhood **neurodevelopmental delay**.
- **No safe level** (CDC). No routine testing, but **assess risk factors in every pregnancy** (Ch 10).

Mercury

- **Congenital Minamata disease** (Japan, 1950s): methylmercury → disturbed neuronal division and migration → microcephaly, intellectual disability, choreoathetosis, motor delay.
- Main source is **large fish**. **Methylmercury is not removed by cooking**.
- **Avoid**: king mackerel, marlin, orange roughy, **shark, swordfish, tilefish**, bigeye tuna.

16. Psychiatric Medications

- **Antipsychotics: none is teratogenic**. Neonates may show transient extrapyramidal signs and withdrawal (agitation, tone changes, tremor, somnolence, respiratory and feeding problems) — FDA alert covers the whole class.
- **Lithium**:
 - **Ebstein anomaly** (apical displacement of the tricuspid valve → severe TR, big right atrium); background **1 in 20,000**.
 - Early registry suggested **3%**; the true attributable risk is only **1–4 per 1000** exposures.
 - **Neonatal lithium toxicity** from exposure near delivery: arrhythmias, hypoglycemia, **nephrogenic DI**, **"floppy infant"** (hypotonia, apnea, bradycardia, cyanosis, poor feeding); may last **2 wk**.
 - → **Reduce or stop the dose 2–3 days before delivery** if possible.
- **SSRIs/SNRIs**: as a class **not major teratogens**.
 - Exception: **paroxetine** → slightly ↑ **cardiac septal defects (ASD/VSD)**, pooled **OR 1.3** (not seen in ~1 million Medicaid pregnancies).
 - **ACOG: avoid paroxetine when planning pregnancy**; consider **fetal echo** after 1st-trimester exposure.
 - **Neonatal behavioral syndrome** in **~25%** with late exposure: jitteriness, irritability, tone changes, feeding problems, vomiting, hypoglycemia, temperature instability, respiratory problems. Mild, lasts **~2 days**.
 - **PPHN**: baseline **2 per 1000**; SSRIs add only **1–2 per 1000**, and cases are not severe.

17. Retinoids

- **Among the most potent human teratogens.** They inhibit **cranial neural-crest cell migration** → **retinoic acid embryopathy**: CNS, face, heart, thymus.
 - **Ventriculomegaly**, facial/cranial bone defects, **microtia/anotia**, micrognathia, cleft palate, **conotruncal heart defects**, **thymic aplasia/hypoplasia**.

Retinoid	Use	Key rule
Isotretinoin (13-cis-retinoic acid)	Nodular acne	High pregnancy loss; up to ⅓ malformed. iPLEDGE (FDA-mandated REMS) for patients, prescribers and pharmacies
Acitretin	Severe psoriasis	Metabolized to etretinate (half-life 120 days ; defects >2 yr after stopping). "Do Your P.A.R.T." → delay conception ≥3 years
Bexarotene	Cutaneous T-cell lymphoma	2 contraceptive methods from 1 month before to 1 month after ; monthly pregnancy tests; male partners use condoms
Topical (tretinoin, isotretinoin, adapalene)	Acne, sun damage	Low absorption; no ↑ malformations in 635 exposures. Tazarotene not recommended (large surface area ≈ oral dose)
Vitamin A	Supplements	Carotenoids (beta-carotene) never teratogenic . Preformed vitamin A (retinol) >10,000 IU/d in the 1st trimester → cranial neural-crest defects. Keep to 3000 IU/d



Figure 8-3. Retinoic acid embryopathy: (A) microtia/anotia with ear canal stenosis; (B) flat, depressed nasal bridge and ocular hypertelorism.

18. Sex Hormones

- **Androgens** → variable virilization of female fetuses and **ambiguous genitalia**.

19. Thalidomide and Analogues

- The most notorious teratogen: malformations in **20%** exposed at **34–50 days menstrual age**.
- **Phocomelia** = absence or underdevelopment of long bones (hands/feet attach to the trunk). Also cardiac, GI, ear and eye defects and other limb reductions. **Up to 40%** of affected newborns die in the neonatal period.
- Marketed **1956–1960** outside the US. Lessons:
 1. **The placenta is not a barrier** to toxins
 2. **Species vary** in susceptibility (no defects in rodents)

3. Timing determines the defect

Defect	Exposure (days postconception)
Upper-limb amelia	24–30
Upper-limb phocomelia	24–33
Lower-limb phocomelia	27–33

- US approval **1999** for **erythema nodosum leprosum** and **multiple myeloma**; restricted via **THALOMID REMS**.
- Analogues **lenalidomide** (MDS, myeloma; limb defects in monkeys) and **pomalidomide** (myeloma, AIDS-related Kaposi sarcoma) → similar restricted programs.

20. Thyroid Medications

- **Methimazole embryopathy** (1–2% of exposed): **2x aplasia cutis congenita** (full-thickness skin defect, usually scalp), **choanal atresia**, **esophageal atresia**.
- **Radioiodine (^{131}I) — contraindicated:**
 - Crosses the placenta; **fetal thyroid concentrates it by 12 wk** → severe/irreversible **fetal and neonatal hypothyroidism** (↓ mental capacity, delayed bone maturation).
 - **Pregnancy test before** treatment; **avoid pregnancy for 6–12 months** after (no ↑ loss or anomalies if conception delayed ≥ 6 months).

21. Warfarin

- Vitamin K antagonist, low molecular weight, long half-life → **crosses the placenta**.
- **Contraindicated except** for women with **mechanical heart valves** at high thromboembolic risk (Ch 52).
- **Warfarin embryopathy** (exposure **6–9 wk**; **6%** risk):
 - **Nasal hypoplasia / depressed nasal bridge**, choanal atresia.
 - **Stippled epiphyses** (femur, humerus, calcanei, distal phalanges).
 - **Dose dependent:** only **1%** with ≤ 5 mg/d.
- **After the 1st trimester: fetal hemorrhage** → abnormal growth and deformation from scarring.
- **~50%** of embryopathy cases have **CNS anomalies**: agenesis of the corpus callosum, **Dandy-Walker**, microphthalmia, optic atrophy → blindness, deafness, developmental delay.



Figure 8-4. Warfarin embryopathy: nasal hypoplasia and depressed nasal bridge on fetal sonography (A) and in the same newborn (B).

22. Tobacco

- **Leading preventable cause of perinatal mortality; $\geq 7\%$** smoke in pregnancy.
- Smoke contains nicotine, cotinine, cyanide, thiocyanate, **CO**, cadmium, lead, hydrocarbons → vasoactive and hypoxic effects.
- **Not a major teratogen**, but:
 - **Orofacial clefts up to 1.5x** (~1 per 2000 attributable).
 - **Poland sequence 2x** (vascular disruption of one side of the chest and arm).
 - **Limb reduction defects** (+25%); small ↑ cardiac defects (dose related).
- **Fetal growth** (most common complication; dose dependent):
 - Babies **~200 g lighter; LBW 2x; FGR 2–3x**.
 - Smoking causes **15% of LBW**. Secondhand smoke also ↑ LBW.
- Also: **preterm birth, PROM, placenta previa, abruption; $\sim 1/3$ of SIDS**; childhood asthma and obesity.
- **All nicotine crosses the placenta; none is safe** — including vaping (fetal brain and lung effects; **EVALI** in adults) and hookah.
- **Cessation**: strongly encourage; avoid secondhand smoke and nicotine-replacement products; **1-800-QUITNOW**; USPSTF: behavioral interventions from the first visit.

23. Herbal Medicinal Products

- **30–60%** of pregnant women use herbal remedies. They are **not FDA-regulated**; identity, quantity and purity are unknown → **discourage use**.

Table 8-5. Pharmacological Actions and Adverse Effects of Some Herbal Medicines

Herb and Common Name	Potential Pharmacological Effects	Concerns
Aloe (oral ingestion)	Smooth-muscle stimulant	May cause uterine contractions
Almond oil (topical)	Smooth-muscle stimulant	Increase in preterm births
Black cohosh	Smooth-muscle stimulant	Causes uterine contractions; also has an estrogenic compound
Blue cohosh	Smooth-muscle stimulant	Causes uterine contractions; contains compounds teratogenic in multiple animal species
Echinacea: <i>purple coneflower root</i>	Activates cell-mediated immunity	Allergic reactions; decreases immunosuppressant effectiveness; possible immunosuppression with long-term use
Ephedra: <i>ma huang</i>	Direct and indirect sympathomimetic; tachycardia and hypertension	Hypertension, arrhythmias, myocardial ischemia; stroke; depletes endogenous catecholamines; life-threatening interaction with monoamine oxidase inhibitors
Evening primrose oil	Contains linoleic acids, a prostaglandin precursor	Possible complications if used for labor induction
Garlic: <i>ajo</i>	Inhibits platelet aggregation; increased fibrinolysis; antihypertensive activity	Risk of bleeding when combined with other platelet aggregation inhibitors
Ginger	Cyclooxygenase inhibitor, thromboxane synthetase inhibitor; lowers blood glucose	Increased risk of bleeding; hypoglycemia
Ginkgo biloba	Anticoagulant	Risk of bleeding; interferes with monoamine oxidase inhibitors
Ginseng	Lowers blood glucose; inhibition of platelet aggregation	Hypoglycemia; hypertension; risk of bleeding
Kava: <i>awa, intoxicating pepper, kawa</i>	Sedation, anxiolysis	Sedation; tolerance and withdrawal
Licorice (glycyrrhizin)	Inhibits cortisol and prostaglandin metabolism	Increased risk of preterm birth
Raspberry leaf	Smooth-muscle stimulant	Increased risk of cesarean when used in late pregnancy or in labor
Valerian: <i>all heal, garden heliotrope, vandal root</i>	Sedation	Sedation; hepatotoxicity; benzodiazepine-like acute withdrawal
Yohimbe	—	Hypertension, arrhythmias

Data from Ang-Lee, 2001; Facchinetti, 2012; Hall, 2012; Munoz Balbontin, 2019; Nordeng, 2011; Strandberg, 2002; Wiesner, 2017.

24. Drugs of Abuse

- Outcomes are confounded by poor health, malnutrition, infection, **polysubstance use**, contaminants (lead, cyanide, pesticides) and by **alcohol and tobacco**.

Drug	Teratogenic?	Main problems
Cocaine	Conflicting (cleft palate, cardiac, urinary tract reported)	Vasoconstriction/hypertension → maternal stroke/cerebral hemorrhage, MI, abruption ; FGR, preterm birth; behavioral and cognitive problems in children
Marijuana	Not a major teratogen, but crosses the placenta; endocannabinoids guide brain development	Most common recreational drug (~4%). ↓ attention, visual problem solving. Preterm/LBW only with concomitant tobacco . No evidence it helps nausea (yet ~70% of dispensaries recommended it). Avoid — ACOG: may be as serious as tobacco or alcohol
Methamphetamine	No	SGA , developmental and behavioral problems; hypertension, abruption, preterm birth, stillbirth
Opioids	Not major (slight ↑ spina bifida, gastroschisis, cardiac in NBDPS)	Heroin → preterm birth, abruption, FGR, fetal death (repeated withdrawal). Neonatal abstinence syndrome in 40–90%
Phencyclidine (PCP)	No	>½ of newborns have withdrawal (tremor, jitteriness, irritability)
Toluene (paint/glue solvent)	Toluene embryopathy — like FAS	Growth deficiency, microcephaly, midface hypoplasia, short palpebral fissures, wide nasal bridge; up to 40% developmental delay

Opioid-use disorder management

- **Neonatal abstinence syndrome:** CNS irritability → **seizures** if untreated, tachypnea, apnea, poor feeding, failure to thrive. Monitor with a scoring system (**Finnegan**); treat severe cases with opioids.
- **ACOG: opioid-agonist maintenance therapy** — **buprenorphine** (office-based) or **methadone** (licensed program), with multidisciplinary care.
- **Withdrawal from methadone in pregnancy is discouraged** (high relapse). Parkland offers **inpatient controlled methadone taper** to women who decline maintenance.

Pearl: The small birth-defect risk with opioid maintenance is far outweighed by the risks of uncontrolled opioid use.

Quick-Recall: Teratogen → Classic Effect

Agent	Classic defect / effect	Critical fact
Alcohol	FAS: short palpebral fissures, thin upper lip, smooth philtrum; growth ↓; CNS	No safe amount
Valproate	NTDs (4%) , 9% major malformations; ↓ IQ	Highest-risk AED
Phenytoin / older AEDs	Fetal hydantoin syndrome (hypertelorism, distal phalanx/nail hypoplasia)	2× risk
Carbamazepine / phenobarbital / topiramate	Clefts, cardiac, NTDs	2× / 3× / 4×
Levetiracetam, lamotrigine	No ↑ risk	Preferred AEDs
ACEi / ARB	Fetal renal hypoperfusion → anuria, oligohydramnios, calvarial maldevelopment, pulmonary hypoplasia, limb contractures	2nd–3rd trimester; ARB 30% vs ACEi 3%
Fluconazole (high dose)	Antley-Bixler-like phenotype	400–800 mg/d chronic
Indomethacin/NSAIDs	Ductus constriction, oligohydramnios	Avoid ≥20 wk; >72 h
Tetracycline	Yellow-brown deciduous teeth	After 25 wk; doxycycline preferred
Nitrofurantoin, sulfonamides	Small ↑ cleft / HLHS; EA/CDH	Avoid in G6PD deficiency
Methotrexate	Cloverleaf skull, craniosynostosis, conotruncal heart	≥10 mg/wk at 8–10 wk
Cyclophosphamide	Skeletal, limb, cleft palate, eye	18% major anomaly
Trastuzumab	Oligohydramnios sequence	Within 7 months before conception
Mycophenolate	Microtia, ear canal atresia, clefts, coloboma	30% defects + 30% loss
Lithium	Ebstein anomaly ; neonatal toxicity (floppy infant, nephrogenic DI)	1–4/1000; stop 2–3 days pre-delivery
Paroxetine	ASD/VSD	OR 1.3; fetal echo
SSRIs (late)	Neonatal behavioral syndrome (25%), PPHN	PPHN +1–2/1000
Isotretinoin / retinoids	Microtia/anotia, conotruncal heart, thymic hypoplasia, CNS	Up to ⅓ malformed; iPLEDGE
Thalidomide	Phocomelia	20% if exposed day 34–50 menstrual
Methimazole	Aplasia cutis , choanal/esophageal atresia	1–2%
Radioiodine	Fetal hypothyroidism	Fetal thyroid uptake by 12 wk
Warfarin	Nasal hypoplasia, stippled epiphyses ; CNS (50%)	6–9 wk; 6% (1% if ≤5 mg/d)
Corticosteroids	Cleft lip/palate	3 per 1000
Tobacco	FGR (–200 g), clefts, Poland sequence	15% of LBW; ⅓ SIDS
Toluene	FAS-like embryopathy	40% developmental delay
Methylmercury	Minamata disease (microcephaly, intellectual disability, choreoathetosis)	Avoid shark, swordfish, king mackerel, tilefish

Key Numbers to Memorize

- Major anomalies **2–3%** of births; background major anomaly risk **~3%**; medications cause **<1%** (0.1% in Utah) of defects; **80%** unknown cause
- Preimplantation "all or none" **2 wk**; embryonic period **2–8 wk postconception**
- Teratogenicity: **RR ≥3.0**, ≥2 studies, or **≥3 cases** for a rare exposure/defect; case-control OR must be **>3–4**

- Low-risk teratogen **<10 per 1000** exposures
- FAS exposure: **≥6 drinks/wk × ≥2 wk** or **≥3 drinks/occasion × ≥2**; growth/HC **≤10th percentile**; neurobehavior **>1.5 SD** below mean
- Valproate **9%** malformations, **4%** NTD; levetiracetam **2%**
- ARB fetopathy **30%**, ACEi **3%** after 20 wk
- Fluconazole high-dose **400–800 mg/d**; NSAIDs avoid **≥20 wk**, ductal risk **>72 h**; aspirin safe **≤100 mg/d**
- Tetracycline teeth after **25 wk**; ribavirin half-life **12 days**, contraception **6 months** after
- Methotrexate **≥10 mg/wk, 8–10 wk**; cyclophosphamide **18%**; trastuzumab **7 months** preconception
- Mycophenolate **30% + 30%**; isotretinoin **up to 1/3**; acitretin wait **3 yr**; vitamin A **>10,000 IU/d** harmful
- Thalidomide **20%**, days **34–50** menstrual; up to **40%** neonatal death
- Lithium Ebstein **1–4/1000** (background 1/20,000); stop **2–3 days** pre-delivery
- Warfarin **6–9 wk, 6%** (1% if ≤5 mg/d); CNS anomalies **~50%**
- Radioiodine: fetal thyroid uptake **12 wk**; avoid pregnancy **6–12 months**
- Smoking: **–200 g**, LBW **2×**, FGR **2–3×**, **15%** of LBW; neonatal abstinence **40–90%**